Remark: $p(x), q(x) \in \mathbb{F}[x]$, $T: V \to V$

① The operators $p(T), q(T): V \to V$

satisfy $p(T) \circ q(T) = q(T) \circ p(T)$

(Recall: in general, $T \circ S \neq S \circ T$)

E.g.: let $p(x) = x+1$, $q(x) = 3x^2 + 4x + 5$

$$p(T) \circ q(T) = (T + I)(3T \circ T + 4T + 5)$$

$$= (T + I)(3T^2 + 4T + 5)$$

$$= 3T^3 + 7T^2 + 9T + 5I$$

$$= r(T),$$

where $r(x) = p(x) \cdot q(x)$

$$= 3x^2 + 7x^2 + 9x + 5$$

② For $\dim(V) < \infty$, $m_T(x)$ always exists (存在)

and $m_T(x)$ is unique

Proof: Suppose that

$p_1(x), p_2(x)$ are polys, satisfying

the properties of minimal poly,

i.e. $\begin{cases} p_i(T) = \underline{0}_{V \to V} & (1) \\ p_i(x) \text{ is monic} & (2) \\ p_i(x) \text{ is of the smallest possible degree} & (3) \end{cases}$

(接上)

(WTS: $p_1(x) = p_2(x)$)

Suppose that $p_1(x) \neq p_2(x)$

(3) $\Rightarrow \deg(p_1) = \deg(p_2)$

(2) $\Rightarrow t(x) = p_1(x) - p_2(x)$ is nonzero

and $\deg(t) < \deg(p_1)$

Remark! $t(T) \equiv p_1(T) - p_2(T) \overset{(1)}{=} \underline{0}_{V\to V} - \underline{0}_{V\to V} = \underline{0}_{V\to V}$

Conclusion:

Let $q(x) = \frac{1}{\alpha} t(x)$, α is the leading coeff. of $t(x)$

Then $q(x)$ satisfies (1)-(2)

and it has

③ Suppose $f(x) \in \mathbb{F}[x]$

$$f(T) = \underline{0}_{V\to V} \Rightarrow \begin{cases} \deg(m_T) \le \deg(f) \\ m_T(x) \mid f(x) \end{cases}$$

Proof: apply division $f(x) / m_T(x)$?

$$f(x) = q(x)\, m_T(x) + r(x) \quad ?$$

$$\begin{cases} \deg(r) < \deg(m_T) \quad ? \\ \text{Then } f(T) = q(T)\, m_T(T) + r(T) \\ \qquad \downarrow = \underline{0}_{V\to V} \qquad \downarrow = \underline{0}_{V\to V} \\ \Rightarrow r(T) = \underline{0}_{V\to V} \end{cases}$$

? By minimality of $\deg(m_T)$, $r(x) = 0$?

and hence $f(x) = q(x)\, m_T(x)$

(Primary Decomposition)

Theorem | : (Use of $m_T(x)$)

$T: V \to V$ is diagonalizable

$\Leftrightarrow$ $m_T(x) = (x-\mu_1)\cdots(x-\mu_k)$

for distinct $\mu_1, \cdots, \mu_k$

Theorem 2 (Cauchy-Hamilton):

(Way to find $m_T(x)$) $\chi_T(T) = \underline{0}_{V\to V} \Leftrightarrow m_T(x) \mid \chi_T(x)$

(Thm 3) (?)

E.g.: (1) $T: \mathbb{R}^4 \to \mathbb{R}^4$, $T(\underline{x}) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \underline{x}$

Then $\chi_T(x) = (x-1)^2(x-2)^2$ (?) ← (by MAT 2040)

By Cauchy-Hamilton Theorem,

$m_T(x)$ can only be

$(x-1)^2(x-2)^2$ | $(x-1)^2(x-2)$, $(x-2)^2(x-1)$ | $(x-1)^2$, $(x-1)(x-2)$, $(x-2)^2$ | $(x-1)$, $(x-2)$

deg 4 | deg 3 | deg 2 | deg 1

↓ (Thm 2) ↓ (Thm 2) ↓ (Thm 2)

↓ Thm 1

$\underline{T}$ is not diagonalizable (?)

E.g.: (2) $T: \mathbb{R}^2 \to \mathbb{R}^2$, $T(\underline{x}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \underline{x}$

(MAT2040) $\Rightarrow \chi_T(x) = x^2 + 1$

① (C.-H.) $\Rightarrow m_T(x) = x^2 + 1$

(Thm. 1) $\Rightarrow$ T is not diagonalizable

(2') $T: \mathbb{C}^2 \to \mathbb{C}^2$, $T(\underline{x}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \underline{x}$

(MAT 2040) $\Rightarrow \chi_T(x) = (x+i)(x-i)$

(C.-H.) $\Rightarrow m_T(x) = \begin{cases} x+i & \text{②} \\ x-i & \text{②} \\ (x+i)(x-i) & \text{①} \end{cases}$

(Thm. 2) $\Rightarrow$ T is diag. over $\mathbb{C}$

Triangularizability & C.-H. Thm.

Def.: $T: V \to V$ is l.-t., $\dim(V) = n < \infty$

T is diag. $\Leftarrow$ $\exists$ basis B of V,

(stronger) s.t. $T_{BB} = \begin{pmatrix} \lambda_1 & 0 & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$

T is triang. (Δ-able) $\Leftarrow$ $\exists$ ~ B ~,

s.t. $T_{BB} = \begin{pmatrix} \lambda_1 & & & * \\ 0 & \ddots & & \\ \vdots & \ddots & \ddots & \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$

Remark: $T: V \to V$ is diag.

$\Leftrightarrow$ $\exists$ basis B of V, s.t. $T_{BB} = \begin{pmatrix} \lambda_1 & 0 & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$

$\Leftrightarrow$ $T(\underline{v_i}) = \underbrace{\underline{0} + \cdots + \underline{0}}_{i-1 \uparrow} + \lambda_i \underline{v_i} + \underbrace{\underline{0} + \cdots + \underline{0}}_{n-i \uparrow}$

$\Leftrightarrow$ $\underline{v_i}$ are e.vectors of T with e.value λ_i

$\Leftrightarrow$ B is a basis of e.vectors.

↘ (def. of diag. in MAT2040

Theorem: $T: V \to V$ is Δ-able

$\iff$

$\chi_T(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_n)$

(Here, $\lambda_1, \cdots, \lambda_n$ are not necessarily distinct)

E.g. $T(\underline{x}) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \underline{x}$ is Δ-able

$\Uparrow$

$\chi_T(x) = (x-1)^2(x-2)^2$

$= (x-1)(x-1)(x-2)(x-2)$

Prop.1: T is Δ-able $\Rightarrow \chi_T(x) = (-1)^n(x-\lambda_1)\cdots(x-\lambda_n)$

Proof: Let B be basis of V,

s.t. $T_{BB} = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

Then $\chi_T(x) = \det(T_{BB} - xI)$

$= \det \begin{pmatrix} \lambda_1 - x & & * \\ & \ddots & \\ 0 & & \lambda_n - x \end{pmatrix}$

(MAT2040) $= (a_1 - x)\cdots(\lambda_n - x) = \downarrow$

Prop 2: $T: V \to V$ is α-able $\iff \chi_T(x) = (x-\lambda_1)\cdots(x-\lambda_n)$

Proof: By $\chi_T(x) = (x-\lambda_1)\cdots(x-\lambda_n)$ & MAT 2040, ①

$\exists \underline{w} \in V \setminus \{\underline{0}\}$ is an e.vector of T
with e.value λ_1

Consider $W := \mathrm{Span}\{\underline{w}\} \leq V$

Then W is T-invariant subspace ①

Take $\alpha \underline{w} \in W$

Then $T(\alpha \underline{w}) = (\alpha \cdot \lambda_1)\underline{w} \in W$

$\Rightarrow T(W) \subseteq W$

(HW6) → Then ① $\begin{cases} T|_W : W \to W \\ \tilde{T} : V/W \to V/W \end{cases}$

and $\exists$ bases $\begin{cases} \mathcal{A} \text{ of } V \\ \mathcal{C} = \{\underline{w}\} \text{ of } W \\ \mathcal{D} \text{ of } V/W \end{cases}$,

s.t. $T_{\mathcal{A}\mathcal{A}} = \left(\begin{array}{c|c} (T|_W)_{\mathcal{C}\mathcal{C}} & * \\ \hline 0 & \tilde{T}_{\mathcal{D}\mathcal{D}} \end{array}\right)$

(1×1) ①

$(n-1)\times(n-1)$